Dataset details

Dataset name and source

Name: Ames Housing (ameshousing, OpenML ID 43926). It is associated with Dean De Cock's Ames, Iowa housing dataset and was published for data-science education. Source/link: OpenML dataset search, ID 43926 (https://www.openml.org/search?id=43926&sort=runs&type=data).

Data format: comma-separated UTF-8 CSV at data/AmesHousing.csv, with one property sale per row and columns for sale price, property characteristics, and neighborhood. The inspected local snapshot contains 2,930 rows and 81 columns.

Purpose in this project: supervised regression learns historical sale-price patterns; property-row documents and metadata support semantic/lexical retrieval of comparable sales. The retrieved records ground the estimate and the UI's evidence section.

Local snapshot profile

The checked-in local file is data/AmesHousing.csv. Its inspected snapshot contains 2,930 rows and 81 columns. The target SalePrice has no missing values in this file; the observed minimum is $12,789, median $160,000, and maximum $755,000. YearBuilt spans 1872-2010 and Neighborhood has 28 distinct values. All model input columns used by the current training configuration were present and had no missing values in this snapshot. The dataset as a whole contains missing values in other descriptive columns.

The upstream OpenML dataset description and this repository's local CSV are separate provenance facts. The application trains and builds retrieval from the local file, and scripts/preparedata.py can fetch and normalize the OpenML source when preparing a fresh copy.

Key fields

Target: SalePrice (historical sale price in US dollars).

Field	Project use
SalePrice	Supervised target; historical sale price in US dollars.
OverallQual, OverallCond	Overall property quality and condition ratings.
GrLivArea, TotalBsmtSF, LotArea	Living area, basement area, and lot area in square feet.
GarageCars, GarageArea	Garage capacity and garage area.
YearBuilt, YearRemodAdd	Construction and remodeling years.
FullBath, BedroomAbvGr, TotRmsAbvGrd	Bathroom, bedroom, and room counts.
Neighborhood	Location filtering, comparable context, and historical neighborhood comparison.
GarageYrBlt	Optional model feature only if supplied by a dataset variant; absent in the current local CSV.

The configured numeric model features are OverallQual, OverallCond, GrLivArea, TotalBsmtSF, GarageCars, GarageArea, YearBuilt, YearRemodAdd, FullBath, BedroomAbvGr, TotRmsAbvGrd, and LotArea. Neighborhood is used for comparable filtering and location evidence, not as a numeric training feature.

Preprocessing performed

scripts/preparedata.py loads OpenML ID 43926 and performs these transformations:

1. Renames source columns to canonical application names (for example SalePrice to SalePrice, GrLivArea to GrLivArea).
2. Converts ordered quality labels to ordinal values 1 through 10.
3. Writes the normalized frame to data/AmesHousing.csv.

During model fitting, scripts/trainmodel.py converts configured model inputs to numeric values, calculates medians from training data, and fills missing input values with those medians. It evaluates configured Random Forest and Extra Trees candidates on an 80/20 split (randomstate=42), selects by lowest MAE, then refits the selected model on all usable records. The model bundle stores its feature names, medians, estimator, and holdout metrics. The current local file has no missing values in the target or configured model input columns; other descriptive columns do contain missing values.

The canonical property document builder converts each row into one compact text document; it does not split a row into multiple chunks. The embedding/index build stores that text and the structured fields as vector and metadata records in ChromaDB. Training and retrieval are historical supervised/search tasks; neither establishes causal effects.

The retrieval index derives property documents and metadata from this same dataset. Local sentence-transformer embeddings are normalized and persisted in ChromaDB. BM25Okapi provides lexical ranking; RRF combines it with semantic ranking.

Limits and intended use

These are historical Ames, Iowa transactions, not current listings or a nationally representative market sample. Values reflect the source dataset’s collection period and should be treated as model estimates, not appraisals, guarantees, or causal claims. Neighborhood and comparable evidence are specific to the available historical records. Retraining from another dataset snapshot can change model scores, predictions, and retrieval results.